



Da Vinci Xi Surgical System In-Service Guide: Surgeon

Software Version: da Vinci OS4 v11

Welcome to Phase II of the da Vinci® Technology Training Pathway. This document is designed to be a facilitator guide for conducting hands-on training for the surgeon during the da Vinci Xi® system overview in-service. The in-service follows the operating room work flow that occurs with use of the da Vinci Xi system. If the surgeon was previously trained on another da Vinci platform, reinforce that while da Vinci controls are similar between systems, setup and use vary due to the mechanics and new features of the da Vinci Xi system.

For reference to the system user manual, go to www.intuitive.com. Updates may be provided electronically.

INTUITIVE

Participant Preparation

Determine if the participants have completed the recommended online training on Intuitive Learning as indicated on the da Vinci Technology Training Pathway. Any activity should be demonstrated by the indicated participant. For discussions, tailor your facilitation style (Q&A vs. didactic) based on the participants' level of preparation. As the facilitator, read all notes, warnings and cautions.

In-Service Participant Roles	Participant Name
Surgeon(s)	
First assistant(s)	
Sterile OR staff	
Non-sterile staff	

Facilitator Instructions

The facilitator may play any role not represented by an in-service participant.

Equipment Preparation

✓	#	Da Vinci Xi Equipment
	1	Da Vinci Xi system
	1	Da Vinci Skills Model
	4	8 mm da Vinci Xi cannulas
	4	5 – 8 mm cannula seals
	4	5 - 12 mm Universal Seal
	1	8 mm blunt obturator
	1	8 mm bladeless optical obturator
	4	Patient cart arm drapes
	1	Column drape

✓	#	Da Vinci Xi Equipment
	1	Instrument release kit (IRK)
	1	8 mm 0° endoscope
	1	8 mm 30° endoscope
	1	Monopolar curved scissor with tip cover
	1	Fenestrated bipolar forceps (training if available)
	1	12 mm 3rd party trocar
	1	Non-energy grasper (training if available)
	1	Da Vinci Xi monopolar instrument cable
	1	Da Vinci Xi bipolar instrument cables

✓	#	3rd Party Equipment
	1	USB flash drive
	1	Patient Neutral Pad
	1	12 mm 3rd party trocar
	1	Dry-erase marker
	3	3-0 RB-1 Suture
	2	0-Vicryl, CT-1 Suture

✓	#	3rd Party Equipment
	1	Laparoscopic monopolar instrument
	1	Laparoscopic bipolar instrument
	1	Ruler
	1	Gauze

✓	Da Vinci Xi System Setup
	Connect all system components
	Power ON the da Vinci Xi system in integrated mode
	Connect Insert endoscope connectors into the holders on the vision cart
	Install instrument and cannula mount sterile adapters
	Install column drape or magnet

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1 System Component Overview

Facilitator note Begin with a Q&A to allow participants to summarize key features and benefits of each major system component. Guide the discussion around value delivered by the features.

Learning objective

Summarize the top 3 features and benefits of each system component.

System component overview discussion



Key features of the surgeon console

3D viewer – enhanced vision of anatomy

Hand controls – intuitive motion translated into precise, fine, tremor filtered movements at instrument tips

Ergonomic design – increases ergonomic comfort for surgeon during surgery



Key features of the patient cart

Boom – adjustable support structure that promotes flexibility in positioning patient cart arms over surgical workspace

Universal arms – allow assignment and re-configuration of endoscope and instruments to desired ports during a surgical procedure, thereby enabling better tissue access and visualization

Helm touchpad – enables non-sterile OR staff to select patient cart setting and adjust boom height/position, improving system setup efficiency



Key features of the vision cart

Touchscreen monitor – provides high quality view of surgical image and allows non-sterile OR staff to adjust vision settings and troubleshoot faults

ERBE VIO® dV Generator – integrated with the system, this electrosurgical unit (ESU) indicates location of installed energy instruments and displays energy effect settings at the surgeon console, increasing surgeon autonomy

Endoscope controller – provides a high-intensity light source to illuminate surgical site and houses electronics that promote efficient endoscope setup

Intuitive Hub (optional) – A single platform that accesses Intuitive's digital ecosystem in the OR that allows capture of procedural audio/video, combined with da Vinci® data, to create content that can be used to collaborate with patients and peers

2 Patient Cart Overview

Facilitator note During this activity, facilitate participant interaction with the boom and arms to build competency in these manipulations.

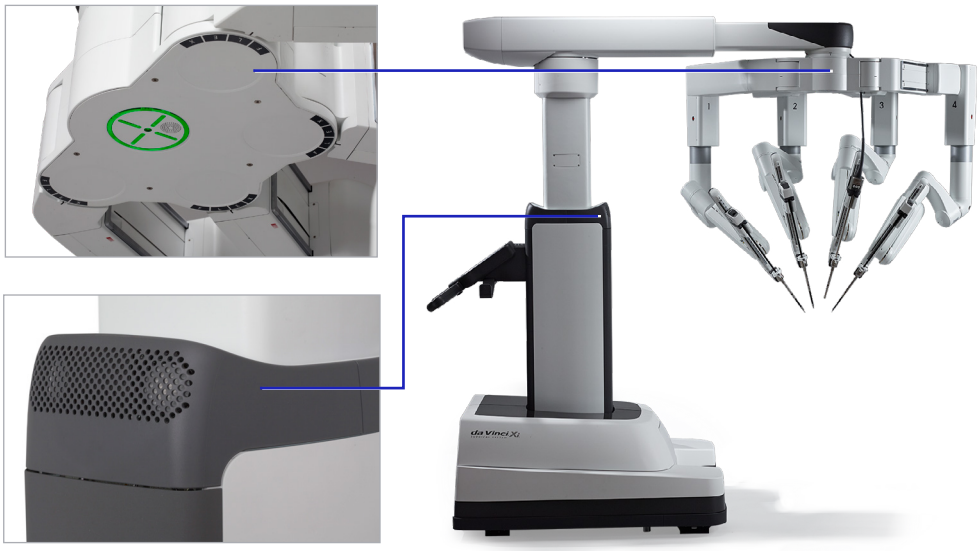
Patient Cart Arms Interaction

Learning objectives

- Identify patient cart components
- Manipulate arms accurately, with single button press and smooth arm movements
- Interpret boom and arm LEDs associated with arm joint activations

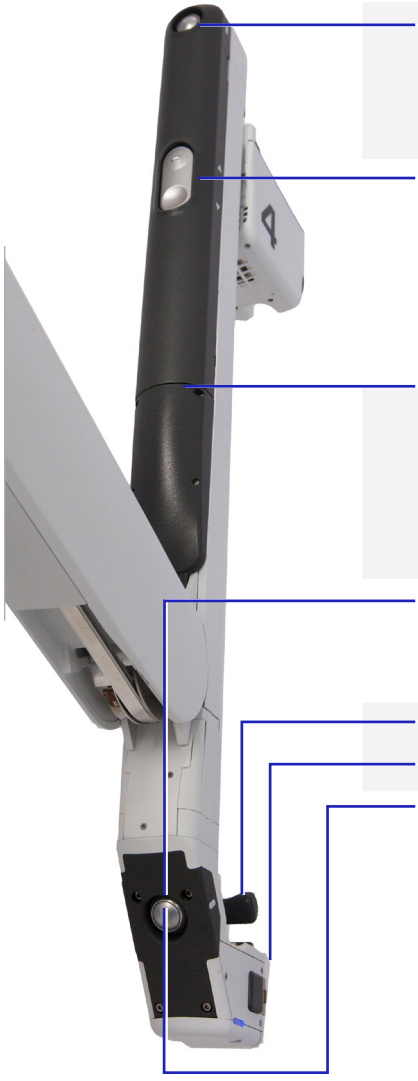
Patient cart components discussion

- Identify patient cart base, column, boom, patient-side speakers and patient-side microphone
- Identify arms numbered 1-4 from left to right: arms universally accept instruments or endoscopes
- Identify instrument carriage, insertion axis, cannula mount, and cannula mount lever



Patient cart overview

Patient cart arms activity



Patient cart arm	OR Application
• Clutch one arm - press and hold or quick click • Instrument clutch and carriage LED blinks blue	• Insert instruments under direct visualization • Adjust arm position without moving port/ cannula
• Raise (▲) and lower (▼) the patient clearance joints on an arm using the patient clearance buttons • Instrument arm LEDs pulse blue	• Easily increase patient clearance as needed during surgery, without removing instruments • Increase internal instrument reach as needed beyond the level of ports, without having to remove instruments
• Grab and move arms individually • Instrument clutch, carriage and port clutch LEDs blink blue	• Make large movements to position undocked arms individually during draping • Move individual undocked arm up and away from patient at end of procedure or for conversion to open or laparoscopic surgery
• Port clutch one arm • Port clutch LEDs blinks blue	• Release tension at port sites • Adjust flex joints to manage arm spacing
• Cannula mount lever • Cannula mount	• Attach cannulas
• Raise and lower boom using extended port clutch - Press and hold the port clutch button on any arm and move it to its upper or lower limit - After the third beep, the arm will move toward the limit at which arm was held • Port clutch LEDs blink blue, boom LED blinks green	• Sterilely adjust boom height to avoid collisions or to access initial port for docking
• Rotate the boom using the boom rotation button on arm 1 or arm 4 • Boom LED blinks green	• Sterilely rotate the boom during mid-procedure re-dock to access a different target anatomy, without relocating the patient cart base

Patient cart overview

Patient cart LED discussion

- Solid blue: System is operating properly
- Blinking blue: Arm is clutched
- Pulsing blue: The arm is performing an activity which requires the user to wait for a brief period
- Blinking green: Guided Tool Change

Patient Cart Helm Interactions

Learning objectives

- Identify patient cart helm components
- Adjust boom height and voice prompt settings
- Make Guided Setup procedure selections

Patient cart components discussion

- Patient cart touchpad

  OR Application: Make Guided Setup selections, set boom height limit

- Power button and battery level indicator

  OR Application: Use the patient cart during a power emergency

- Boom position joystick

  OR Application: Manually adjust boom extension, pivot and rotation

- Boom height joystick

  OR Application: Manually adjust boom height

- Emergency STOP button

  OR Application: Stop da Vinci® surgery during an emergency

- Cart drive enable switches

  OR Application: Move and position the patient cart

Patient cart overview

Helm settings activity

1. Settings tab → Select "Reduced" to turn off non-critical Guided Setup voice prompts.

   OR Application: "Reduced" turns off tutorial voice prompts. If Reduced is selected, a message appears informing that tutorial voice prompts are turned off.

2. Move volume slider to adjust volume of patient cart speaker audio and voice prompts.

   OR Application: The patient cart speaker audio can be increased in a noisy OR to enable clear two-way communication between the console surgeon and patient-side staff.

3. Settings tab → Select "On" to turn on boom height limit.
4. Select "Modify" and follow on-screen instructions to modify boom height limit for the room where training is taking place. Select "Save."

   OR Application: In operating rooms with low ceilings or overhead obstacles, setting a boom height limit will avoid external collisions when rolling up the patient cart to the patient.

Guided Setup

- Provides step-by-step instructions with audible voice prompts, to successfully prepare, position and dock the patient cart to the patient
- Instructions for the sterile surgeon and patient-side staff are displayed on the vision cart touchscreen. Instructions for the non-sterile circulating staff are shown on the helm touchpad.

Guided Setup activity

1. Select anatomy
2. Select patient cart location: side of the patient that the cart will approach from.

OR Application: Select the side applicable for the current OR in which training is being held. Once selected, the touchpad image should match room setup.



Warning: Be sure to understand the function of the Patient Cart joystick and the use of the setup controls to prevent patient injury.

Learning check

1. When would you use each button on the patient cart arm?
2. What are the sub-components of the patient cart helm?

3 Instruments and Accessories Overview

Facilitator note Facilitate participant interaction with the cannulas, cannula seals, obturators, instruments and endoscopes.

Learning objective

Interact with and set up sterile accessories, endoscopes, and instruments for a da Vinci® procedure.

Accessories overview activity

Identify the following:

- Da Vinci cannulas
  - 8 mm reusable, available in 10 cm (standard) and 15 cm (long) lengths
  - Cannula fin
  - Remote center markings on both cannula lengths
    - The remote center is the pivot point around which the arm moves
    - Positioning remote center in the muscle layer of the body wall helps minimize the amount of force the system may exert on the body wall during surgery
- Cannula seals
  - 5-8 mm disposable seals
  - 5-12 mm Universal Seal
  - Universally used with both 8 mm instruments and endoscopes
  - All have inlet for use with insufflation or smoke evacuation as needed
  - Can also be used with 5 mm laparoscopic instruments
  - Place the cannula seal into the cannula bowl and push down until you hear an audible click
  - To remove seal, squeeze the latches towards each other and pull cannula seal out of cannula
- Da Vinci obturators
  - 8 mm blunt reusable in 10 cm and 15 cm lengths
  - 8 mm bladeless optical disposable in 10 cm and 15 cm lengths
  - Attach obturator (blunt or bladeless) to cannula assembly, ensuring that it snaps into the cannula with an audible click to indicate full engagement

Note For additional instructions on da Vinci bladeless obturator assembly, setup, and use, refer to the Instrument and Accessories User Manual.

Instruments and accessories overview

Accessories overview activity continued

- Da Vinci Hasson Cone
  - 8 mm (reusable)
  - 12 mm & stapler (reusable)



8 mm



12 mm

Note For additional instructions on da Vinci Hasson Cone assembly, setup, and use. Refer to the Instrument and Accessories User Manual Addendum.

Note Use only instruments and accessories approved by Intuitive Surgical. System compatibility with non-approved instrumentation cannot be guaranteed. Improper assembly or disassembly of instruments or accessories can lead to tissue injuries.

Note For additional information about the EndoWrist Suction Irrigator, refer to the da Vinci Xi EndoWrist Suction Irrigator In-Service Guide.

Warning: Never use a double cannula (for example, an 8 mm da Vinci Xi cannula installed through the da Vinci Xi 12 mm & Stapler cannula) on the da Vinci Xi System or the da Vinci X System. This may cause instrument damage or an imbalance of the remote center, and potential injury to the patient.

Warning: Reprocessing or reuse of products intended for single use only may result in degraded instrument performance or loss of functionality, and in exposure to viral, bacterial, fungal, or prionic pathogens.

Caution: Da Vinci instruments must be used with the appropriate size and type Intuitive cannula, because tissue injury or instrument damage could result.

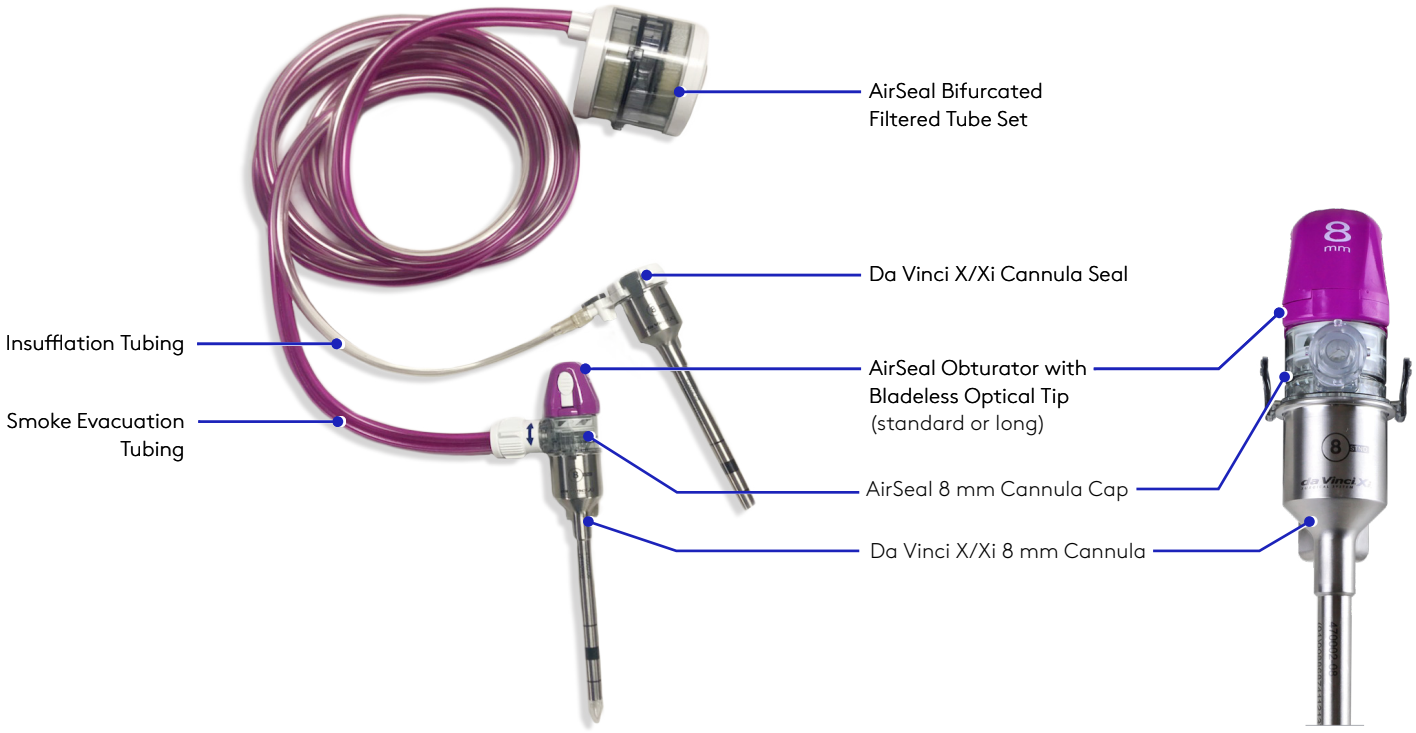
Third-party accessories discussion

The ConMed AirSeal® 8 mm cannula cap and obturator with bladeless optical tip comes in standard length and long length, and is compatible with the standard length da Vinci X/Xi 8 mm cannula and the long length da Vinci X/Xi 8mm cannula. The da Vinci X/Xi 8 mm blunt or bladeless obturator (optical) is not compatible with the AirSeal Cannula Cap.

The ConMed AirSeal Bifurcated Filtered Tube Set with Activated Charcoal Filter is compatible with the following da Vinci X/Xi Cannula seals:

- 5 - 8 mm Cannula Seal
- 12 mm & Stapler Cannula Seal
- 5 - 12 mm Universal Seal

The image below represents a complete setup of ConMed AirSeal products with da Vinci X® and da Vinci Xi®accessories. For additional instructions on ConMed AirSeal accessories, including assembly information, refer to the ConMed AirSeal Accessories user manual.



Caution: Unintended instrument tip motion can occur when the instrument carriage and accessories connected to the corresponding cannula bowl collide, resulting in patient injury.

Note Using third party accessories connected to the cannula may result in reduced range of motion of instruments due to interference between accessories and instrument carriage.

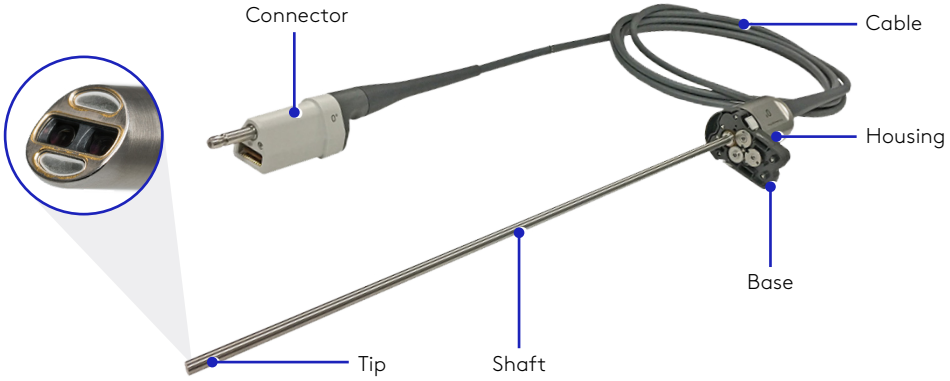
Instruments and accessories overview

Endoscope Plus overview activity

3D high definition endoscope

- For 0° and 30° endoscope, identify the following:
  - Integrated base, connector, housing, shaft and tip
  - Angled tip of 30° scope

See the Endoscope Plus Training Guide for information regarding physically differentiating endoscope plus from previous generation endoscopes.



Note Inspect da Vinci Xi endoscopes (0° and the 30°) for deposits or damage to lenses, cable or integrated connector before and after each use.

- Identify buttons on endoscope housing to prevent patient tissue injury from unintentionally turning off the endoscope during a procedure:

1. Right/Left eye swap and Targeting: Identify vision/fogging issues in one eye of the endoscope or press and hold to activate the targeting feature
2. Take Photo: Capture a still image of the endoscope view
3. Illuminator on/off: Turn on the illuminator during laparoscopic use or turn off the illuminator sterilely upon endoscope removal from the patient cart arm



Warning: Avoid direct interaction with the illuminator to prevent burn injury or infrared radiation injury to the user.

Endoscope Plus overview activity continued

Plug-in the 30° endoscope integrated connector to the endoscope controller on the vision cart.

OR Application

- In preparation for a da Vinci case, connect endoscope ~30 minutes prior to surgery
- Ensure endoscope is securely connected and ready for use

Note It is physically possible to install more than one endoscope on the Patient Cart arms. However, only the endoscope connected to the Endoscope Controller sends video to the 3D viewer and touchscreen monitor. If more than one endoscope is installed on the arms, a system message appears indicating that only one endoscope will send video.

Identify Endoscope Controller status LEDs:

- Powering up (blue, flashing)
- Powered up and ready (blue, steady)
- Failure (red)
- No power (LED off)

Practice orienting the 0° and 30° scopes for laparoscopic use, with the base always facing up:

- For the 0° scope: buttons facing up
- For the 30° scope: align up or down arrow on the housing with the hash mark on the base
- Take care not to exert excess force on the endoscope shaft during laparoscopic use
  - Excessive force could cause bending of the endoscope shaft, especially when placed between two hard surfaces (ex: ribs, model)

OR Application: Da Vinci Xi endoscopes can be used laparoscopically during port placement and as needed.

Warning: Improper use or handling of any components that could get hot could lead to burn injuries.

EndoWrist instrument overview activity

- Identify 8 mm da Vinci Xi EndoWrist® instrument components:
  - Tip, wrist, shaft, housing, release buttons, instrument release socket, maximum use indicator, flush ports, and discs
- Demonstrate/mimic the instrument's 7 degrees of freedom:
  - Insertion, external pitch, external yaw, roll, grip, internal pitch, and internal yaw
- Discuss instrument lives:
  - Most have 10-18 lives
  - Clip appliers have 100 fires
  - Training instruments have 30 lives or 100 fires
  - An instrument life is used as soon as the instrument goes into following mode from the surgeon console
  - If an instrument is expired, the maximum use indicator on the side of the housing will turn red

Monopolar curved scissors setup activity

Monopolar curved scissors (MCS) tip cover

The tip cover is a single use accessory that is intended to provide insulation over a section of the monopolar curved scissors so that energy is only available at the tip.

- The MCS must always be used with the tip cover
- The tip cover should be inspected for damage if collisions occur during the procedure
- If the tip cover has been damaged in any way, remove the MCS to replace the tip cover, to prevent arcing through the defect and to prevent energy delivery to unintended tissue resulting in patient injury
- Improper application can lead to components coming off in the patient



Tip cover installation



Tip cover properly installed

Install the MCS tip cover following these steps:

1. Inspect for damage before installation – do not use a torn tip cover accessory
2. Close MCS blades and straighten the wrist
3. Grasp tip cover with blue installation tool and slide it onto the distal end until it stops
4. Remove installation tool and retain it
5. Check that the orange surface is fully covered, but the tip cover is not installed over the shaft
    a. The tip cover is not properly installed if any part of the orange surface is visible



Tip cover not properly installed



Tip cover over-installed

Warning: Do not use another instrument to clean the Tip Cover Accessory to prevent energy delivery to unintended tissue resulting in patient injury.

Learning check

1. How do you orient the 30° and 0° endoscopes laparoscopically?
2. How many lives do most EndoWrist instruments have?

4 Port Placement and Docking

Facilitator notes

- Demonstrate port placement and facilitate participants' practice of placing ports for multiple surgical workspaces.
- Have participants practice steps specific to their roles in the OR. Play the other roles as required.

Port Placement

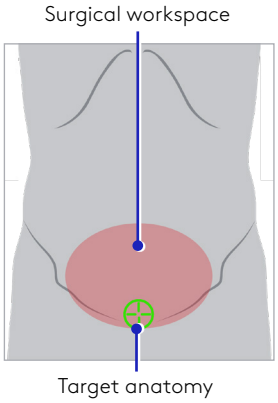
Learning objectives

- Describe port placement philosophy and general guidelines to follow when placing ports for a da Vinci® case
- Practice placing ports for example surgical workspaces relevant to the surgical specialties that the participants support
- Follow Guided Setup for preparing, positioning, and docking the patient cart
- Setup for a 3-Arm procedure

Surgical workspace activity

Define surgical workspace

1. Area in the patient that all instruments will need to reach to successfully complete the procedure
2. Mark on the da Vinci Skills Model for a representative procedure



Target anatomy activity

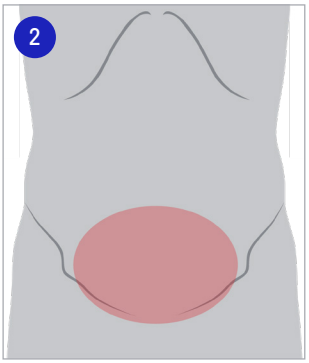
Define target anatomy

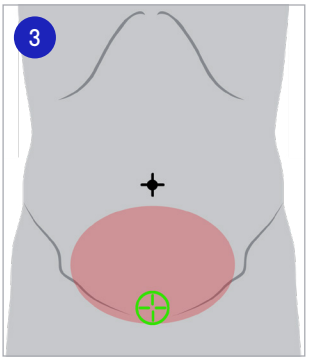
1. From the center of the surgical workspace, the target anatomy is the farthest point required to reach within the workspace
   a. This is not necessarily the location of pathology
2. Mark on the da Vinci Skills Model for a representative procedure

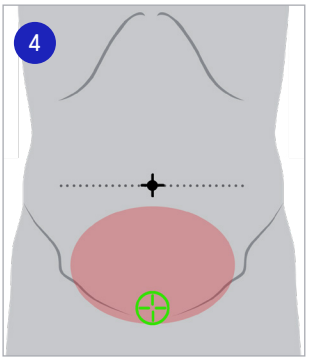


Port marking activity

1. Before marking port locations, insufflate the abdomen.
2. Identify the surgical workspace:
   a. If two quadrants or less, proceed to identifying the target anatomy.
   b. If greater than two quadrants, consider dual docking.
3. Identify a pelvic target anatomy, and mark the endoscope port 10 – 20 cm from target anatomy.
   a. If the workspace comprises two quadrants or less, the endoscope port will be at or beyond the edge of the workspace, ensuring adequate visualization.
   b. If the desired port location is greater than 20 cm from the target, consider dual docking.
4. A linear port placement is recommended to minimize arm-to-arm interferences.

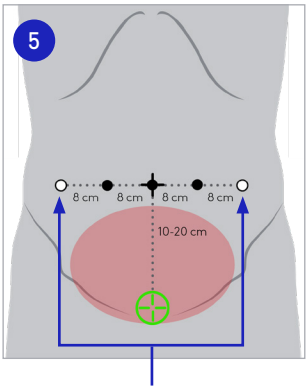






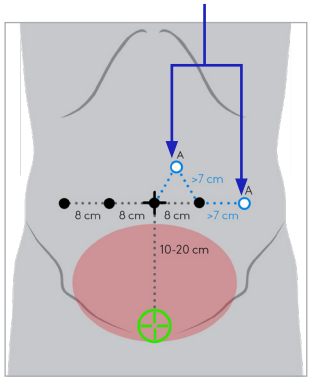
Port marking activity continued

5. For instrument arm ports, measure from the initial endoscope port perpendicular to the axis between the target anatomy and the initial endoscope port.
   a. Mark ports 6 – 10 cm (8 cm recommended) from initial endoscope port in a line.
      - Ports can be a minimum of 4 cm apart in certain cases (e.g. procedures utilizing a small surgical workspace and requiring minimal outer arm motion, such as mitral valve repair).
   b. Maintain 2 cm or more between ports and any bony prominences.
   c. Do not place ports between other ports and the target anatomy.
6. Mark the retraction arm port on either side of the patient, 6 – 10 cm (8 cm recommended) from the nearest da Vinci port, and in line with all other da Vinci ports.
   a. Consider location of internal anatomy to be retracted.
   b. Consider placing on same side as monopolar instrument (allows grasper use in both hands).
   c. Consider placing arm opposite of the assistant.



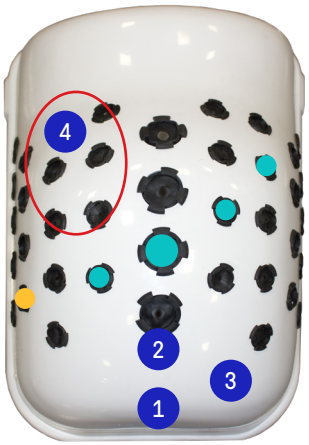
Retraction port options

Assistant port options



Port marking activity continued

7. Consider port hopping potential when placing ports.
   a. Placing 4 ports allows switching between 2 right hands/left hands, changing vantage point.
   b. Placing 5 ports allows switching between two right hands and two left hands, while maintaining vantage point. The additional port can be used as an assistant port.
8. Mark the assistant port lateral to or triangulated between the da Vinci ports. Ensure the assistant port is at least 7 cm away from all other ports and has a clear path to the target anatomy.
   a. Use long-length laparoscopic instruments for greater reach and access.
   b. Do not place the assistant port between da Vinci ports and the target anatomy.
9. Mark ports for a second representative procedure relevant to your surgical specialty.
   a. For an upper right quadrant procedure, mark port placement as shown. The red outline represents the surgical workspace, and "4" represents the target anatomy.

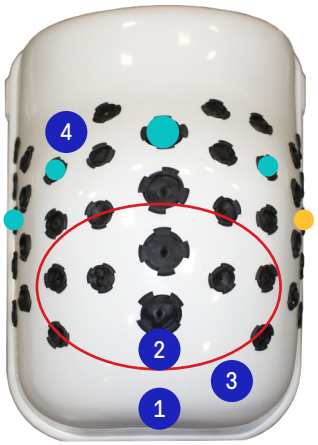


Port location:
Upper right quadrant procedure

Port placement activity

If needed, re-mark ports for a pelvic procedure. Using the endoscope to provide direct visualization, place the instrument and accessory ports into the abdominal model as shown. The red outline represents the surgical workspace, and "2" represents the target anatomy.

1. Insert the endoscope port into the previously marked location
2. Insert the endoscope, and visually confirm the surgical workspace and target anatomy before placing additional instrument ports
3. Use the endoscope to provide direct visualization for placing the remaining ports
   a. Align the remote center of each cannula with the inside edge of the abdominal model, corresponding to the muscle layer of the abdominal wall
4. Insert the remaining instrument and accessory ports in the locations marked



Port location:
Pelvic procedure

Patient positioning discussion

- Position the patient for a representative procedure at surgeon's discretion, following hospital protocol
- Once the patient cart is docked to cannulas in the patient, the patient cannot be repositioned

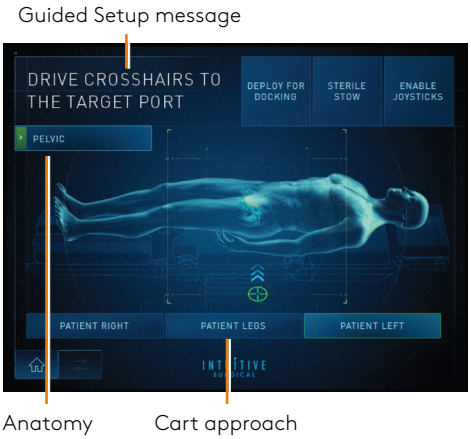
Docking

Guided Setup activity: Deploy for docking

1. Visually verify the anatomy and patient cart location selections on the vision cart touchscreen Guided Setup tab
2. Boom deployment position varies based on the anatomy and patient cart location selected at the helm
3. Ensure there are no obstructions overhead or in front of the patient cart
4. Following Guided Setup messages on the vision cart touchscreen and audible voice prompts, guide circulating staff to deploy the boom for docking
    a. Monitor boom movement to avoid collision with equipment or the patient
5. The boom will raise, extend, and pivot, orienting the patient cart arms based on the anatomy and approach that were previously selected at the patient cart helm

Guided Setup activity: Drive laser lines to the endoscope port

Using room landmarks for directions instead of “left” or “right”, direct the circulating staff to drive the patient cart towards the patient until the green positioning laser lines projected on to the floor are within 5 cm of the initial endoscope port.



Port placement and docking

Guided Setup activity: Dock the arm you will use for the endoscope

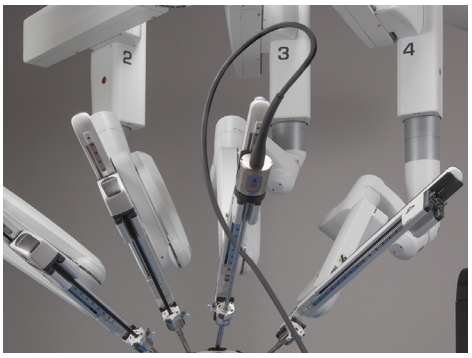
Select the arm that will act as the initial endoscope arm for the procedure and dock it to the port that will initially be used
for the endoscope:

1. Lower the arm down to the cannula using the port clutch button
2. Angle cannula toward the cannula mount
3. Hold down the cannula mount lever with thumb while holding down the port clutch button
4. Insert the cannula fin into the cannula mount and release the cannula mount lever
5. When the cannula is properly seated, the lever will return to the upright position
    a. If it is not properly seated, the lever will remain down and a message will appear on the touchscreen over the associated arm
    b. Once a cannula is docked to an arm, the helm touchpad will indicate “Surgery in Progress”, show anatomy and cart location selections, and disable helm positioning functions
    c. Patient cart stabilization feet lower, locking cart into place

Warning: Do not move the Patient Cart while one or more arms are docked to the patient, to prevent harm to the user or patient.

Guided Setup activity: Install the scope

1. Insert the 30° endoscope into one of the undocked ports to check remote center of the initial endoscope port and adjust as necessary, which will help minimize the amount of force the system may exert on the body wall during surgery
    a. Ensuring correct placement of the remote center before docking an arm helps minimize the amount of force the system may exert on the body wall during surgery
2. Install the 30° endoscope onto the arm
    a. The endoscope will rotate to the position in which it was last used laparoscopically, to maintain the vantage point
    b. Do not place the endoscope cable in any pinch points
3. Secure endoscope cable between endoscope and insertion axis of the arm
4. Place endoscope tip in cannula, then install the base onto the sterile adapter until fully seated
5. Observe that the illuminator turns on automatically upon installation



Correctly secured endoscope cable

Port placement and docking

Facilitator note

To facilitate incomplete targeting activities, set a boom height limit of "90", and increase table height. Instruct participants to attempt targeting to complete the below activity. After practice, ensure that targeting is successful before proceeding with the remainder of Guided Setup.

Guided Setup activity: Targeting

1. Quick click the instrument clutch button, and insert the endoscope into the Skills Drills Model.
2. Point the endoscope at the target anatomy, and lock the endoscope in place with a quick click of the instrument clutch button.
   a. Level the horizon of the endoscope to ensure proper boom orientation during targeting.
3. The targeting button is used to orient and align the patient cart boom to the center of the surgical workspace. Aligning the initial endoscope arm with the target anatomy enables optimal orientation of all patient cart arms over the surgical workspace.
4. Use one hand to stabilize the cannula remote center at the model's body wall to prevent patient injury. With the other hand, press and hold the targeting button.
5. If targeting does not complete, remove the endoscope and undock the arm. Attempt one of the following to troubleshoot incomplete targeting:
   a. If possible, follow the vision cart touchscreen message to "lower the operating table for improved sterile clearance."
   b. Raise or disable the boom height limit from the settings tab of the helm touchpad.
6. Reattempt targeting, and observe the following:
   a. The boom will rotate to align with the principal working direction.
   b. The boom will pivot to center over the initial endoscope port.
   c. The boom height will adjust based on the height of the docked arm to ensure that all arms will be able to reach the remaining cannulas, as well as provide adequate clearance between the top of the patient cart arms and the boom.
7. After targeting completes, use the positioning laser lines as a guide to confirm that the laser lines transect the ports. These lines remain on for a brief period of time after targeting has completed, or until a second cannula is docked.
8. Using the port clutch button, flex the endoscope arm to bring it in line with the center of the workspace (i.e. along the positioning laser that points towards target anatomy).
   a. When using the port clutch button to reposition an arm, always stabilize the cannula at the body wall.

Warning: Be sure to understand the function of the targeting feature, including supporting the cannula while targeting, to prevent patient or user injury.

Manual boom orientation discussion

If targeting does not complete after lowering the operating table and raising the boom height, manually position the boom and arms using the following guidelines:

- Rotate the boom towards the target anatomy
- Ensure all arms can reach cannulas for docking
- Ensure there is enough space between the boom and arms to remove instruments in a sterile fashion

Warning: Undock the instrument arms prior to moving the patient, table, or Patient Cart to prevent patient harm.

Facilitator note

To facilitate the remaining docking activities, ensure that targeting is successful before continuing.

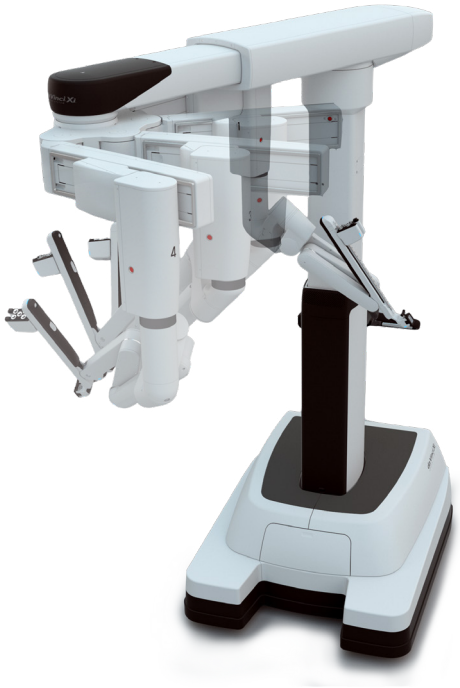
Guided Setup Activity: Docking and System Alignment

1. Dock the remaining arms as previously instructed.
2. After the last arm is docked, the Guided Setup tab disappears.
3. Arm spacing is adjusted to maintain parallel configuration of the arms because:
    a. Insertion axes of the da Vinci Xi arms do not telescope. External and internal arm-to-arm interferences are minimized in a parallel setup.
4. Starting from the arm closest to the initial endoscope arm, flex the remaining arms towards the initial endoscope arm, maintaining about a fist's space between them.
5. Check system setup spacing and ensure there is no tension at any port.
    a. To release tension, stabilize the cannula at the body wall with one hand and press the port clutch button with the other hand.
6. If maximum instrument reach is anticipated in a large surgical workspace, lower the patient clearance joint to about a fist's space to the patient. This joint of each arm can be adjusted for either patient clearance or instrument reach.

Port placement and docking

3-Arm procedure setup activity

1. Drape all 4 patient cart arms and dock the arms to be used for the procedure, including performing targeting
2. Collapse the arm to be stowed onto itself using the instrument clutch button
3. Use the port clutch button to fully rotate the arm inward towards the other arms, until it is facing the column in the opposite direction of the other arms
4. Raise the arm to its upper most limit
5. Check that LEDs on the arm turn off and the 3D viewer displays " arm stowed"
6. Once the procedure is over, be sure to un-stow the arm and move it back to its normal orientation before moving the boom



Learning check

1. Define target anatomy and surgical workspace in the context of a da Vinci Xi procedure.
2. What purpose does targeting achieve?

5 ERBE VIO dV Generator

Facilitator notes

- Confirm that the ERBE VIO® dV generator is powered on
- Connect patient neutral pad

ERBE VIO dV Generator

Learning objectives

Interpret and interact with the ERBE VIO dV (VIO dV) generator user interface and complete typical electrosurgical unit (ESU) setup tasks:

- Describe energy settings on the VIO dV generator interface
- Navigate error logs for reference and troubleshooting

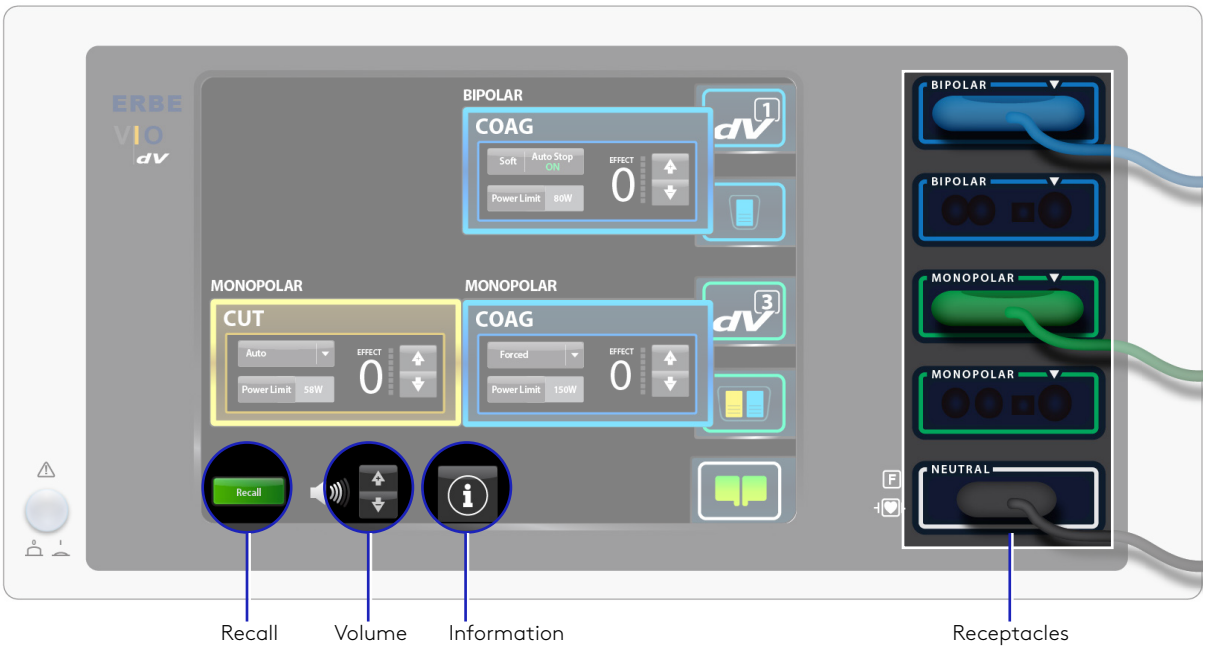
VIO dV overview activity

The VIO dV is an integrated electrosurgical unit (ESU) that can be used with da Vinci Xi® and handheld instruments.

- Identify receptacles for da Vinci® instruments and handheld instruments
- Open the "i" (information) menu, scroll down and review the characteristics and effect range for each mode
  - Close the menu when finished

Note: Energy settings can be adjusted from the surgeon console

Caution: Ensure that the energy is delivered from the intended generator to the intended instrument, to prevent patient or user injury.



ERBE VIO dV generator

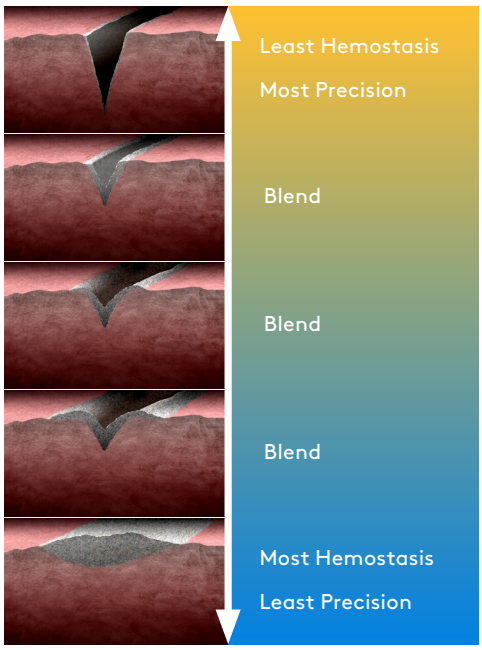
Energy settings discussion

- Auto Cut is the default mode for monopolar Cut
- Swift is the default mode for monopolar Coag

Caution: Using the Surgeon Console to change settings on one monopolar instrument changes settings for all connected monopolar instruments. Changing settings on one bipolar instrument changes settings for all connected bipolar instruments. This can result in an unintended energy level being delivered to the patient, causing tissue damage.

VIO dV monopolar modes

	Tissue Effect	Pedal	Effect Range
AUTO CUT	Least Hemostasis / Most Precision	CUT	1-8
DRY CUT	Blend	CUT	1-8
CLASSIC COAG	Blend	COAG	1-2
SWIFT COAG	Blend	COAG	1-8
FORCED COAG	Most Hemostasis / Least Precision	COAG	1-4



VIO dV bipolar mode

BIPOLAR COAG	Soft Coag: Lower voltages, carbonization of the tissue is prevented, adhesion of the electrode to the tissue is very much reduced AUTO STOP: Ends activation of BIPOLAR SOFT COAG automatically before the tissue adheres to the instrument	COAG	1-8

ERBE VIO dV generator

Effect settings discussion

- Effect setting changes relative amount of coagulation or cutting effect for the selected mode, and can increase the coagulation zone.
- Set the effect as low as possible relative to the required surgical effect. If unsure what effect to use, begin with effect 1.



Effect

Power limit discussion

In addition to adjusting mode and effect settings, users can set and adjust power limit settings. This enables you to continue to influence smoke development, carbonization, cutting behavior and coagulation behavior in fine increments for all monopolar and bipolar instruments. A power limit activity will be performed later in the in-service.

Error log activity

Error log tab displays the 20 most recent error messages. This log can be used for quick reference and troubleshooting.

1. Touch the “i” (information) button located at the bottom of the ERBE VIO dV touchscreen
2. The Help Screen will appear and by default, will display the “Error Log” tab
3. Tap the “Back” button to return to the main page



Learning check

Describe the monopolar Coag modes.

6 Intraoperative Setup and Use

Learning objectives

- Practice insertion, removal and exchanging of endoscopes and instruments
- Manage and troubleshoot endoscope fogging
- To prevent breach in sterility practice handling the instruments, endoscopes and accessories

Endoscope fogging discussion

- Technique for intraoperative cleaning:
  - Remove the endoscope from the arm
  - Wipe the tip with moistened sterile gauze
  - Briefly submerge the endoscope tip in warm water: temperature less than 55°C (131°F)
  - Wipe both the tip and shaft dry with sterile gauze
  - Use sterile gauze to clean the lumen of the cannula seal prior to installing the endoscope
- When the endoscope is outside the patient and waiting to be used, it can be placed in a scope warming device
  - Do not use warming devices that apply temperatures greater than 49°C (120°F) for extended periods of time (longer than 5 minutes)
- Use heated insufflation and check that insufflation/smoke evacuation tubing is not connected to endoscope cannula

Instrument insertion activity

For this activity, install a bipolar instrument in arm 1, MCS in arm 3, and grasper in arm 4.

1. Before inserting an instrument, clutch the endoscope and view the cannula in which the instrument will be installed to check and adjust remote center position
2. Straighten the instrument wrist and close the instrument tips
3. When using energy instruments, attach energy cables to the instruments before installing onto patient cart arm
4. Place instrument tip in the cannula then install housing onto sterile adapter until fully seated
5. Observe audible feedback and blue pulsing carriage LED during information download
6. Solid blue carriage LED and an audible tone indicate installation completion
7. Check icons on the assistant monitor to ensure that the instrument is engaged

Instrument insertion activity

8. Quick click the instrument arm and endoscope arm clutch buttons (blinking blue carriage LEDs)
    a. Observe off-screen indicator with associated arm number on the vision cart touchscreen
        - Yellow and black for active instruments
        - Gray and black for inactive instruments
9. Visualizing the cannula tip with the 30° scope, manipulate both arms to introduce instrument until its tip is just visible, then guide it towards the target anatomy
10. Quick click the instrument arm clutch button again to lock the instrument into place
    a. Do not move any instrument that is not under direct visualization
    b. To prevent tissue injury, ensure that there is adequate room for instruments to move inside the patient, as unexpected motion can occur causing instruments to collide

Practice instrument insertion with a facilitator or another in-service participant manipulating the endoscope.

Practice instrument insertion independently manipulating both endoscope and instrument arms.

Instrument removal activity

Warning: Use proper instrument removal techniques (for example, straightening instrument wrist, freeing instrument jaws of patient anatomy) to avoid damaging tissue or the instrument upon removal.

Practice read-back technique between console surgeon and patient side assistant to remove an instrument:

1. Visually confirm that the instrument is free and clear of any patient anatomy
2. If needed, request console surgeon to straighten instrument wrist and slightly open jaws
3. Use read-back technique to confirm which instrument to remove
    a. Identify the instrument by both name and arm number
4. Confirm that you are ready for the instrument to be removed
5. Squeeze release buttons on instrument housing and pull the instrument straight up and out without using excessive force to prevent instrument damage and patient harm

Warning: Be aware of the risk of damaging the drapes during instrument removal, to prevent a critical patient infection.

Warning: Insert and exchange instruments under vision to prevent tissue injury.

Note: During intraoperative use, do not clean tissue off of instruments with other instruments. Have the patient-side assistant remove the instrument and clean off excess debris with moist sterile gauze as necessary.

Guided Tool Change activity

Guided Tool Change is a feature that enables the system to remember the location of the last-installed instrument or endoscope on an arm.

1. Re-install an instrument onto the patient cart and notice the blinking green LEDs
2. The touchscreen will also show a message that states "Advance instrument to return to previous location" above the arm on which Guided Tool Change is active
3. Observing the monitor, and without pressing the instrument clutch button, push the instrument's housing to slowly guide the installed instrument through the cannula until you reach a hard stop, 3 mm short of the previous instrument tip location
   - Do not use excessive force during insertion to prevent patient tissue damage
   - Pressing any clutch button will clear memory and disable Guided Tool Change
   - In the absence of Guided Tool Change, always insert instruments under direct visualization of the endoscope
4. Practice using Guided Tool Change for endoscope-to-endoscope exchanges or endoscope removal and re-insertion to manage intra-operative fogging
5. Repeat steps 1-3 and demonstrate Guided Tool Change with a straight wrist and a bent wrist
6. If the system does not recognize a re-installed instrument, the arm is temporarily disabled and a message appears
   a. Remove the instrument, re-seat the sterile adapter and re-install the instrument
   b. If the issue persists, contact customer service

Port hopping activity

The endoscope or an instrument can be installed on any of the 4 arms as needed.

Note: Pay attention to arm associations and instrument control when the endoscope is port-hopped to prevent inadvertent instrument movement or removal, which could lead to tissue injury.

1. Based on the activities completed in this section, install instruments in the configuration below, under direct visualization:
   a. Arm 1 – Bipolar instrument
   b. Arm 2 – Endoscope
   c. Arm 3 – Monopolar instrument
   d. Arm 4 – Grasper
2. Remove the bipolar instrument and grasper from arms 1 and 4 respectively
3. Use Guided Tool Change to install and insert the grasper in arm 1
4. Remove the monopolar from arm 3 and perform a Guided Tool Change to install it on arm 4
5. Remove the endoscope from arm 2. Install the endoscope on arm 3 (no Guided Tool Change as arm 3 previously had an instrument)
6. Under visualization, install and insert the bipolar on arm 2, guiding it towards the target anatomy (no Guided Tool Change as arm 2 previously had an endoscope)
7. Perform port hopping again, restoring instruments back to their original locations while verbalizing which ones will have Guided Tool Change and which will not

Learning check

1. What are the best practices for managing endoscope fogging?
2. What is Guided Tool Change?

7 Surgeon Console Interaction

Facilitator note Place a moistened piece of gauze in the Skills Model under endoscopic view.

Component Overview

Learning objective

Identify surgeon console components and describe their functions.

Component overview discussion

Stereo viewer with 3D vision

- Ergonomically designed
- Head rest
- Infrared head sensor: prevents movement of the patient cart arms when the surgeon's head is not in the viewer
- Speaker and microphone

Right-side pod

- Power button: turns the da Vinci® system on and off
- Emergency STOP button: creates a recoverable fault and halts system operation

Left-side pod: ergonomic controls

- Stereo viewer height
- Stereo viewer tilt
- Arm rest height
- Footswitch panel depth

Touchpad

- Main system setting and video control interface of the surgeon console

Hand controls

- Follow the surgeon's movements precisely, translating movements to the surgical arms in real time
- Allow autonomous control of the EndoWrist® instruments and endoscope
- Allow a natural range of movement and ergonomic comfort, even during extended procedures

Footswitch panel

- Arm swap pedal
- Endoscope control pedal
- Clutch pedal
- Instrument activation pedals

Dual console option

- A second console allows two surgeons to collaborate during a procedure
- To operate in dual console mode (using a second surgeon console) simply plug the second console's system cable into an available port on the back of the vision cart core
- No other connection is necessary, except to a power outlet on a dedicated circuit

Touchpad Interaction

Learning objectives

- Use the touchpad to log in and setup surgeon console ergonomics
- Describe functionality of the Instrument tab and Settings tab features

Surgeon console setup and login activity

Unlock touchpad: Tap "Unlock" to adjust settings on the touchpad. Note that when the surgeon takes control of instruments, the touchpad will automatically lock.

Log into the applicable user account to use:

- **Intuitive® account:** Use this login if the surgeon has an active Intuitive account. Follow the on-screen prompts to set up an Intuitive account.

  Note: This login option may not be available on all systems or geographies. Contact your sales representative for more information.

- **Console-only account:** Use this login if the surgeon does not have an Intuitive account, so access is to a console-only account. For new and returning users, follow the on-screen prompts to set up a console-only account.

Note For additional information on Changing the Primary Procedure, Changing the Primary Surgeon, Managing Procedures With Another Surgeon Logged in to the Same Case, and Changing Surgeons During a Case, refer to the da Vinci Xi System User Manual the Intuitive Account Login Video or the P11 Key Changes Document.

Surgeon console interaction

Instrument tab discussion

Instrument Tab: Provides status for all four arms and enables quick adjustment of video and audio. In dual console mode, it also enables surgeons to give or take control of the arms. The status and controls associated with each of the arms is displayed on the instrument tab as arm status pods arranged horizontally across the screen.

Arm pod quick settings	OR Application
Endoscope Pod	
• Brightness	Adjust visible light or Firefly® mode background brightness (not illumination)
• Firefly	Activate or deactivate Firefly mode
• Scope angle	Rotate the angle of the 30° scope up or down when a 30° scope is installed on an arm
Instrument Pod	
• Give/Take (dual Console)	"Give" when the user has control and "Take" when the other console has control
• VIO dV settings	Access the VIO dV control panel to adjust settings or view messages for energy instruments
Quick-settings bar	
Messages	Toggle non-critical system messages between visible and hidden
Audio	Mute or enable the surgeon console microphone and adjust the speaker volume
Energy controls	Open the VIO dV control panel or view troubleshooting information
Give All/Take All (dual console)	Give or take of all instruments from one surgeon console to another
Take photo	Capture an image from the endoscope view when an endoscope is connected to the endoscope controller and a USB flash drive is connected to the video processor
TilePro™	Show or hide the TilePro image

Surgeon console interaction

Settings tab discussion

Settings Tab: Allows the surgeon to configure a variety of system settings. The available settings are grouped in the following sections: Image, Display, TilePro, Firefly, Hand Controls, and Account.

Press the icon next to an item on the screen to view further information about that topic, and then press "Close".

Settings tab	OR Application
Image Button	
• Restore to factory defaults	Restore video settings back to their default values
• Sharpness *	The sharpness slider adjusts the sharpness of tissue edges in the displayed image
• Brightness	Adjust brightness of the surgical image
• Color balance	Adjust the color balance of the endoscopic image • Sliding to the left increases the red tones in the image (e.g. when working near fibrous or avascular tissue) • Sliding to the right increases the blue tones in the image (e.g. when working near fatty or vascular tissue)
• Color modes *	The color mode features allow you to change the endoscopic image by adjusting the color and image characteristics • Standard Color mode (default) produces an image similar to open surgery - Tissue differentiation is more pronounced and blood appears redder than in Classic Color mode • Classic Color mode produces an image with a comfortable hue while keeping whites bright - It is ideal for looking at lower or constant contrast scenes (for example, fat, bowel, muscle) - This mode closely matches the "Normal Color mode" available on past Xi endoscopes * Endoscope plus only - See Endoscope Plus Training Guide
• Dynamic contrast	Adjust contrast of endoscopic image while in visible light mode. Increasing contrast will brighten darker regions of the image and darken brighter regions
Display button	
• Digital zoom	• Zoom in on the image without changing the position of the endoscope tip • Zoom has three settings 1x, 2x, 4x
• Haptic zoom	Turn haptic zoom on or off
• Working distance	Adjust to see a fused 3D image at normal or close up distances
• Telestration eye	Toggle display of telestration overlay between left and right eye images • The vision cart touchscreen has a telestration feature that allows non-sterile staff to draw a colored line on the vision cart touchscreen
• Viewer mode	Toggle the 3D viewer between 2D and 3D mode

Surgeon console interaction

Settings tab discussion continued

Settings tab	OR Application
TilePro button	
• TilePro (On/Off)	Turn TilePro display on or off
• TilePro QuickClick	Turn TilePro QuickClick on or off • When it is enabled, the surgeon can switch to TilePro mode with a quick press of the endoscope pedal
• TilePro size	Scale the TilePro image using the buttons • The live endoscopic image scales to fill in the remaining display **Note:** In dual console mode, the console must be assigned all active instruments in order to make TilePro size changes. If a console that is not assigned all active instruments attempts to make a TilePro size change, a system message appears.
• TilePro Viewer mode	Display a 3D TilePro video source in the 3D viewer, when a compatible video source is connected to both the TilePro (L) and (R) connectors
Firefly button	
• Firefly mode	Activate or deactivate Firefly mode touchpad
• Firefly fingerswitch	Activate to use the finger switches to toggle between visible light and Firefly modes 1. Ensure the Firefly Fingerswitch is set to "ON" 2. Press and hold the endoscope pedal down 3. While pressing the endoscope pedal, toggle into and out of Firefly mode by using either of the finger switches **Note:** Surgeon should locate the target anatomy while in visible light mode before administration of ICG and then switch into Firefly mode
• Firefly intensity	Increase the contrast between the fluorescing tissue and non-fluorescing tissue
• Background brightness	Adjust brightness of black and white image in Firefly mode
• Sensitive mode	For more information about sensitive mode see the Sensitive Firefly Training Guide
• Restore to factory defaults	Restore Firefly settings back to default values

Surgeon console interaction

Settings tab discussion continued

Settings tab	OR Application
Hand Controls Button	
• Scaling	Scale the sensitivity of the hand controls in relation to the instruments • Quick (1.5:1) • Normal (2:1) • Fine (3:1) ratios
• Finger clutch	Turn finger clutch on or off
• Hand control assignments	Open the hand control assignments screen, which enables manual association of either hand control with any instrument • Press Auto to apply hand control assignments automatically, or press Manual to switch hand control association of individual instrument arms - When instruments are reassigned, the system takes the instruments out of surgeon control, notifies and prompts user to tap the arm swap pedal • The system does not allow the user to associate more than two arms with one hand control and notifies if attempted: “Maximum of two arms per side”
Account button	
• Restore saved settings	• Return settings to user-saved values • Access the Login tab and tap and hold the "Restore Ergonomic Settings" button - The Restore Saved Settings button is available when a user is logged in to the surgeon console
• Restore factory defaults	Restore to factory default settings
• Import settings	Import settings from another user profile

Surgeon console interaction

Surgical Controls

Learning objective

Use the surgical controls and interpret 3D viewer interface to complete typical surgical tasks.

Surgeon console viewer discussion

Instrument status pods show instrument name, arm number, instrument activation functions and hand control associations. Messages relevant to the installed instruments and endoscope appear over the applicable arm status pods.

Facilitator note

To facilitate the below activity, cover one of the installed instrument tips completely, with folded gauze, paper, or your hand.

Tool association activity

1. Without touching the hand controls, put your head in the 3D viewer
2. Use the tool association icons to identify instrument arm numbers and respective hand control associations
3. Describe which instrument is closer to the endoscope and which one is farther

OR Application

- Tool association icons can be used to infer the depth of inserted instruments. The icon sizes are approximated to the corresponding insertion depths of instruments.
- Sometimes, instrument tips are obstructed by tissue. Tool association icons help you recognize and recover from such a scenario.

Footswitch panel discussion

- Arm swap pedal: tap the pedal quickly with side of foot to switch control to the inactive instrument arm
- Endoscope control pedal: press and hold the pedal to disengage the hand controllers from instrument control and enable control of the endoscope
- Clutch pedal: press and hold the pedal to disengage the hand controls from instrument control and enable repositioning for ergonomic comfort
- Instrument activation control
  - Primary instrument activation pedals are blue, and secondary instrument activation pedals are yellow
  - SmartPedal™ Technology automatically assigns any instrument controlled by the left hand control to the left instrument pedals and any instrument controlled by the right hand control to the right instrument pedals

Surgeon console interaction

Hand controls activity

- Head-in requirement:
  - Always place your fingers in hand controls before placing your head in the stereo viewer
  - Always take your head out of the stereo viewer before taking their hands out of the hand controls to prevent patient tissue damage
- Match grips:
  - Before taking control of instruments, match the position of your grip on the hand controls to the position of the instrument tips
  - Open, close, or slightly rotate the hand controls to match grips
  - Do not match grips for instruments whose tips are not visible in the 3D viewer to prevent serious harm to the patient
- Reposition the hand controls:
  - To reposition both the hand controls at the same time:
    - Press and hold the clutch pedal on the footswitch panel

      - **or** -
    - Pull back on both finger clutches on hand controls to disengage hand controls from the instrument
  - To reposition each hand control independently:
    1. Pull back on the finger clutch to disengage the hand control from the instrument
    2. Reposition the hand control for ergonomic comfort
    3. Release the clutch pedal or finger clutch(es) to re-engage the hand controls with the instruments

Caution: Fully understand the finger clutch capability to avoid unintentional instrument movement and prevent patient tissue damage.

Warning: Do not manipulate instruments that are outside the surgeon's field of view, to prevent patient tissue damage.

Surgeon console interaction

Endoscope control activity

- Press and hold the endoscope control pedal
- Move the hand controls to reposition the image
  - To move the endoscope towards the target anatomy, pull the hand controllers toward you
  - To move the endoscope away from the target anatomy, push the hand controllers away from you
  - To rotate the endoscope, rotate both hand controllers to the left or right (like turning a steering wheel)
  - To move the endoscope from side to side, move both arms together to the left or right
- Haptic zoom: a method to apply digital zoom using the hand controllers, while in camera control mode (digital zoom does not move the endoscope tip)
  - Press and hold the endoscope control pedal
  - To increase digital zoom, push hand controls towards each other (in a clapping motion)
  - To decrease digital zoom, pull hand controls apart from each other (in a reverse clapping motion)

Facilitator note To facilitate the below activity, move one active instrument and one inactive instrument out of endoscope view.

Off-screen indicator activity

1. Observe the off-screen tool indicators for active (yellow and gray) and inactive (black and gray) instruments
2. Move the endoscope in the direction of the indicator to find the instruments
3. Bring the instruments back to the endoscope's original location, keeping them constantly in view to prevent tissue injury

OR Application

- Sometimes, an instrument such as a grasper may become out of view retracting tissue. Before moving endoscope view to a different region within the surgical workspace, use the off-screen tool indicator as a reminder to locate the out-of-view instrument and bring it into view.
- Do not move an instrument that is out of endoscope view.

Surgeon console interaction

ERBE VIO dV Generator

Facilitator note Confirm that you are logged in as a new user, not an existing user.

Learning objectives

- Describe modes and effect settings on the VIO dV touchpad
- Adjust power limit and presets
- Rename and access presets

ERBE VIO dV settings adjustment activity

1. From the surgeon console touchpad, adjust VIO dV bipolar and monopolar effect settings:
   a. Tap the "Energy" button

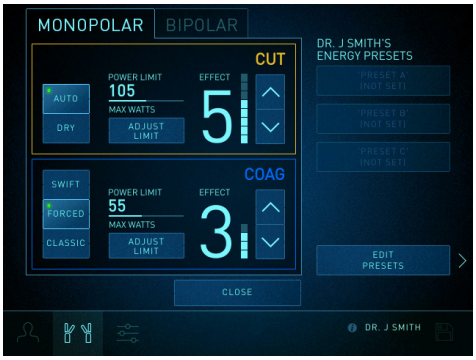
      - OR -

   b. Tap the "Energy Controls" quick-setting to access the VIO dV Settings panel



Surgeon console interaction

Monopolar energy settings activity



Adjust monopolar **cut** mode, effect and power limit settings:

1. Tap the "Mode" button: Auto or Dry
2. Tap the effect up/down arrow
3. Tap the "ADJUST LIMIT" button
4. Drag the slider to make large adjustments
    a. Moving the slider to the right increases the power limit
5. Tap "DEFAULT" to reset to factory default power limit

Adjust monopolar **coag** mode, effect and power limit settings

1. Tap the "Mode" button: Swift, Forced or Classic
2. Tap the effect up/down arrow
3. Drag the slider to make large adjustments
    a. Moving the slider to the right increases the power limit
4. Tap the left/right arrows to make incremental adjustments

Bipolar energy settings activity



Adjust bipolar mode, effect and power limit settings

1. Tap BIPOLAR tab
2. Tap the Effect up/down arrow
3. Tap the "ADJUST LIMIT" button
4. Tap the left/right arrows to make incremental adjustments
5. Tap "AUTOSTOP" button
6. Tap "CLOSE"

Surgeon console interaction

Energy presets discussion

- Users can apply and edit presets only when the VIO dV is turned on and user is logged on to console
- Up to three preset energy settings can be saved to a user account
- Presets can be set to be applied by default at login
- Presets can be given custom names
- From the surgeon console touchpad, presets can be edited, saved and applied
- From the vision cart touchscreen, presets can ONLY be applied

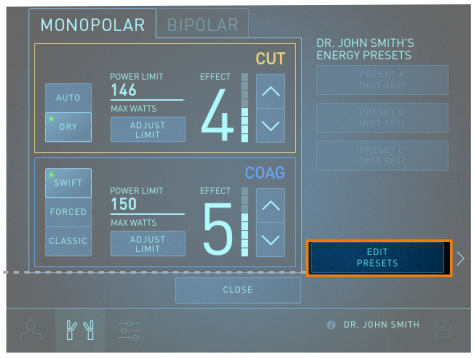


Energy presets activity (from the energy controls button)

1. Tap the “Energy Controls” button
2. Adjust Mode and Effect

   Note: If adjusting Power Limit in addition to Mode and Effect, select the “ADJUST LIMIT” button, adjust all three settings, and then tap the “ENERGY PRESET” button
3. Tap “EDIT PRESETS” button
4. Select “EDIT PRESET B” button from the preset list
5. Tap “SAVE ENERGY SETTINGS”
6. Confirm “SAVE SUCCESSFUL” screen is displayed
7. Tap “CLOSE” button

   Note: Only one preset for an account can be set to be the default at login





Caution: Adjusting energy settings on the touchpad always adjusts the energy settings on the VIO dV immediately.

Rename preset activity

Renaming the Preset is optional

1. Tap the “ENERGY CONTROLS” button or the “ENERGY CONTROLS” quick-setting button
2. Tap the “EDIT PRESET” button
3. Select “EDIT PRESET B”
4. Tap “RENAME” button
5. Type “GYN” using the keypad
6. Tap “SAVE NAME”
7. Tap “CLOSE”



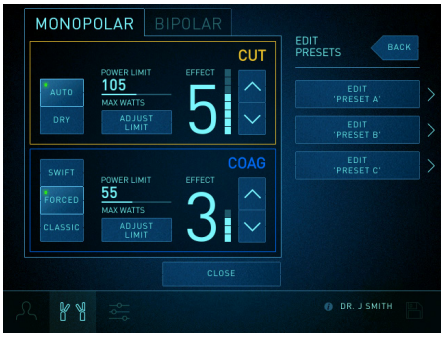
Surgeon console interaction

Apply preset activity

There are three ways to apply presets:

1. From the surgeon console touchpad:
   a. Tap the "ENERGY CONTROLS" button from the Instrument tab
   b. Select "PRESET B" button from the preset list
   c. Tap "APPLY" from the confirmation screen
   d. Tap "CLOSE"
2. From the vision cart touchscreen **single console** setup:
   a. Tap the Settings tab
   b. Tap "ENERGY PRESETS" button
   c. Select a preset from the preset list
3. From the vision cart touchscreen **dual console** setup:
   a. Select Settings tab

      Note: if no preset is applied, the button will read "NO PRESET SELECTED"
   b. Tap a user's name to display that user's presets
   c. Select a preset from the preset list

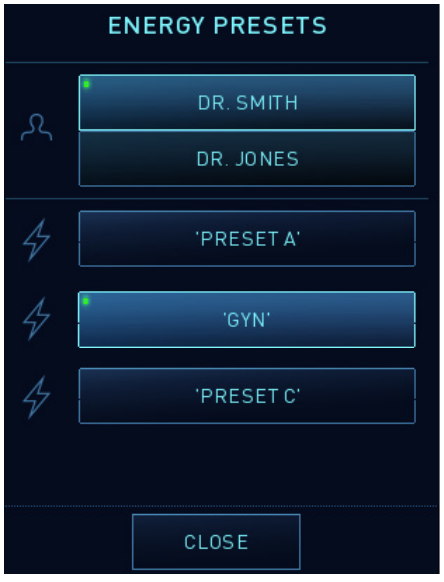




Single console

Bipolar and monopolar settings

1. Using two instruments of same energy type:
   a. Even though there are two monopolar and two bipolar ports on the ERBE VIO dV generator, there is only one set of monopolar settings and one set of bipolar settings on the generator.
   b. If there are two monopolar or two bipolar instruments, a message displays stating that the settings apply to all connected monopolar or bipolar instruments. This message must be acknowledged by tapping "OK" before the settings panel opens to allow adjustment.
2. If a message other than the energy settings display on the instrument arm pod, tap the Display button for troubleshooting guidance:
   a. Check Energy Cord – problem with connection between instrument and VIO dV generator
   b. VIO dV Not Present - VIO dV generator is not powered on



Dual console

Surgeon console interaction

Instrument activation activity

1. Hover your foot over the right bank of energy activation pedals for the monopolar instrument in arm 4: observe green border along the right half of the viewer, and a green highlight around “cut” and “coag” in the arm status area.
2. Press and hold yellow pedal, to activate monopolar “cut”: observe green border and green highlight are replaced by corresponding yellow border and yellow background for “mono cut.”
3. Similarly, activate monopolar coag: observe blue border along right half of viewer, and blue background in the arm status area.
4. Practice activating bipolar energy in arm 1 (bipolar instrument). Grasp gauze with bipolar instrument. Observe green hover and blue fire indicators along left half of the viewer and the instrument arm status area.

Warning: Improper setup of ESU or inadvertent activation of energy instruments could lead to patient harm and may result in energy delivery being unavailable.

Warning: As with any electrosurgical device, it is possible for energy to discharge in an area other than the instrument tip. It is important to exercise caution when using an energized EndoWrist Monopolar Curved Scissors instrument to help avoid unintended contact with tissue adjacent to the area to be cauterized.

Warning: Activation of the incorrect activation pedal could result in unintended instrument function, causing tissue injury.

Warning: Instrument and cannula configurations that are electrically isolated from the body wall have the potential to enable the unintended transfer of monopolar energy.

Surgeon console interaction

Dual Console Interaction

Facilitator note

Plug in second surgeon console's blue system cable into an available port on the vision cart core, and connect its AC power cable to a dedicated circuit. Play the role of the other surgeon to facilitate this activity.

Learning objectives

- Use the surgical controls and interpret 3D viewer interface to complete typical surgical tasks
- Interact with system features and a second surgeon in a dual console setup

Dual console activity: (discussion only, if second console not available)

- Either surgeon can take control of the endoscope arm by pressing the endoscope control pedal at any time
  - Use of the endoscope control pedal by either surgeon causes the instrument arms to go out of surgeon control for both surgeons until the endoscope control pedal is released
- Practice read-back technique between the two surgeons to complete the following:
  - From the Instrument tab, give one arm to surgeon in second console
  - From the Instrument tab, take one arm from the surgeon in the second console
  - From the Instrument tab, select "Give/Take all" to give/take ownership of all instruments
- Give ownership of instruments to the other surgeon console and observe your hand control movements translate into "virtual pointers"
  - Use this feature for clear communication with the surgeon in the other console
    - Clear communication is necessary to avoid inadvertent instrument motion that could lead to tissue injury

Instrument Manipulation

Facilitator notes

- Refer to da Vinci Skills Drills Practicum and Setup Guide
- Install da Vinci Xi EndoWrist needle drivers in arms 1 & 3 so they can reach both ends of rollercoaster
- If using the simulator, facilitate practice of "thread the rings" activity
- Encourage practice of instrument/endoscope manipulation, and regular clutching

Setup and facilitate the following activities:

1. Ring Rollercoaster activity
   a. Alternatively, complete the "thread the rings" exercise on the da Vinci Xi Skills Simulator
2. Suturing activity
   a. Alternatively, complete the "suture sponge 1" activity on the da Vinci Xi Skills Simulator

Learning check

1. What are two ways to activate digital zoom?
2. What is SmartPedal Technology?

8 Troubleshooting

Facilitator note To facilitate the below activity, position the patient cart arms close and facing one another, to trigger external and internal arm interferences.

Learning objectives

- Troubleshoot arm interferences and surgical workspace reach during setup and intraoperative use
- Describe emergency stop, manual instrument grip release and conversion processes, and the conditions under which these are performed
- Interpret vision cart touchscreen interface and patient cart LEDs to manage intraoperative system and generator faults

Arm and Instrument Management

Intraoperative arm management activity

Resolve external interferences between the patient cart arms:

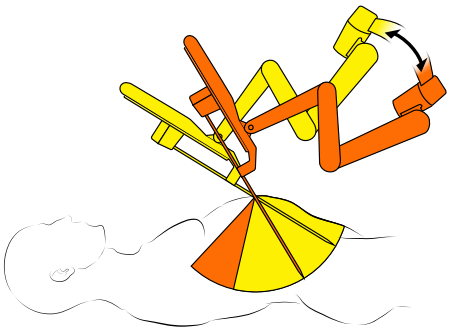
- Adjust flex joint to manage spacing between the front ends of the arms
  1. Remove the instruments from the arms that will be adjusted
  2. Stabilize the cannula while you reposition the arm using the port clutch button
  3. Reinsert the instrument manually under direct visualization
- Adjust patient clearance joint to manage spacing between the back ends of the arms
  1. Have the console surgeon move the tip of the instrument away from tissue
  2. Adjust the patient clearance joint up or down to provide additional spacing
  3. Note that instrument control from the surgeon console is paused when the patient clearance joint is being adjusted
- Create space between patient and patient cart arms by adjusting the patient clearance joint up

Arm and Instrument Management continued

Increase instrument reach activity

Increase reach lateral to the surgical workspace (in line with ports)

1. Starting with the arm closest to the lateral workspace, flex all arms toward the direction where reach is desired, maintaining them parallel
2. When working away from this lateral region, consider flexing the arms to the center of the workspace for greater reach in the rest of the workspace



Increase reach beyond the level of ports

1. Have the console surgeon move the relevant instrument tip away from tissue
2. Adjust the patient clearance joint down to gain additional reach beyond the level of ports

The flex and patient clearance joints are adjusted to increase reach in the desired area during initial setup as well as during intraoperative use of the da Vinci Xi® system. Before adjusting flex joints, remember to remove instruments as this adjustment impacts the ports.

Fault Management

System faults discussion

When a fault occurs, the system determines whether the fault is recoverable or non-recoverable. The system behaves in the following manner when a fault occurs:

- Locks all the patient cart arms
  - In this state the patient cart clutch buttons function normally but the instrument is slightly harder to move and is not able to be left in clutch mode
- Sounds a series of error beeps
- Displays a text message on the monitors to describe the error
- If the fault is arm-specific, the system displays an error icon and the specific arm LED will be amber or red
- If the fault is not arm-specific, all arm LEDs will be amber or red

Recoverable fault activity

A recoverable fault is one that the system can recover from and does not require a system restart in order to resolve the fault. Amber LEDs on the arm/system are associated with a recoverable fault and the fault can be overridden by pressing the “Recover Fault” button. Follow the steps below to prevent patient harm.

1. Trigger a recoverable fault by pressing the “Emergency STOP” button on the patient cart helm or the surgeon console right side pod
2. Read messages on the vision cart touchscreen Fault tab and observe amber LEDs/audible feedback
3. Once the issue is resolved, tap the “Resume Use” button

Note If the fault is arm-specific, the system displays an error icon above that arm on the touchscreen and the specific arm LED will be red or amber depending on the type of fault. A “Disable Arm” option is displayed on the touchscreen and Surgeon Console touchpad. A disabled arm cannot be re-enabled until the system is restarted.

Non-recoverable fault discussion

A non-recoverable system fault is associated with red arm/system LEDs and requires a system restart

Assuming non-recoverable fault, complete the below troubleshooting steps to prevent patient harm:

1. Read messages on surgeon console touchpad and vision cart touchscreen
2. Where possible, leave the instruments and endoscopes installed and inserted in the patient
3. Press the Power button on any system component to power off the system
4. Once powered off, wait for all system power buttons to be lit amber, indicating standby mode
5. Press the Power button on any system component to restart the system

Note During system restart, video is temporarily unavailable at the Surgeon Console view and touchscreen monitor.

Manual Instrument Grip Release

Instrument release kit activity

In the event of a system fault where normal instrument removal is not possible, or when surgeon console control of the instrument is not practical or available, use the Instrument release kit (IRK).

- The Instrument Release Kit can be individually sterile-wrapped, labeled, and placed in the vision cart drawer for easy access.

If the instrument tips are holding tissue, the instrument release wrench allows the patient-side operator to manually open the grips to release tissue – instruction card is attached to wrench. Note that the instrument should not be re-used on an arm after the instrument release wrench has been applied.

1. To prevent patient injury, ensure that the system is in a fault state. If not, press the "Emergency STOP" button on the surgeon console right side pod to trigger a system recoverable fault.
2. Insert the long end of the wrench into the grip release socket on the instrument housing. Using the short end could lead to over-rotating which can cause unintended instrument motion or damage to the manual grip release mechanism, which could lead to tissue injury.
3. Stabilize the housing with one hand while pushing the wrench into the socket to ensure it engages and to prevent patient injury.
4. Turn wrench in the direction of the arrow on the housing approximately 1/4th turn to prevent unintended instrument motion or damage to the manual grip release mechanism, which could lead to tissue injury.
5. Under endoscopic visualization, verify the grips are not grasping tissue.
6. Once the tissue is released from the grips, remove the wrench from the instrument.
7. Squeeze the release buttons on the sides of the instrument housing and remove instrument.
8. Tap "Recover Fault" or "Resume Use" on the touchscreen or touchpad, and restart the system as necessary.
9. Notify Technical Support of the issue. Provide event error code following these steps:
    a. On the vision cart touchscreen, touch Settings > Troubleshooting > Display Event Logs.
    b. The system log appears on the touchscreen. Use the navigation buttons to scroll through the log. Press "Close" to close the log.

Warning: Improper use of the Instrument Release Kit may cause injury to the patient or tissue.

Conversion and Power Loss

Conversion to open and lap/immediate patient access discussion

If a situation arises where conversion to open surgery or immediate access to the patient is required:

1. Remove the instruments from the patient under endoscopic or direct visualization where possible
    a. If needed, use surgeon console control to release the instrument grips
    b. If robotic control cannot be used to release the instrument grips, manually release grips using the Instrument Release Kit to prevent tissue damage when removing an instrument grasped on tissue
2. Remove the endoscope from the patient
3. Disconnect the cannulas from the arms
4. Move the arms away from the patient - If necessary, move patient cart away from the patient to prevent patient harm

Undocking the system activity

Practice undocking the patient cart following the steps below to prevent patient injury:

1. Remove the EndoWrist® instruments (if inserted) from the patient cart arms under direct visualization
2. Remove the endoscope from the patient cart arm
3. Detach the cannulas from the patient cart arms by pushing back the lever and detaching the cannula fin
4. Raise all the arms so they safely clear the patient and compact them as necessary to facilitate roll back
5. If additional clearance is needed, use “extended port clutch” to raise the arm group

Power loss situation discussion

- Battery level indicator at patient cart helm: number of green LEDs reduces as level reduces
- If the LED is red, it indicates that ≤ 10% battery is available and that the battery must be charged immediately
- In a power loss situation where the backup battery is depleted, you may need to manually raise the patient cart stabilization feet and place the patient cart in manual drive, in order to move it away from the patient

Troubleshooting

Power loss situation activity

1. Open the access panel in the front of the patient cart base
2. Follow the instructions in the label inside the panel to place the patient cart into manual drive
3. Make sure that the patient cart is not docked to the patient
    a. If required during power loss, undock the arms and gently move the arms up along their vertical joints, overpowering the resistance to move the arms up and away from the patient
4. Use two people to manually move the patient cart
5. As soon as manual control is no longer needed, move the lever back to the powered position
    a. This restores normal functionality of the system



Learning check

1. Describe system behavior during a system fault
2. Describe the steps to use the instrument release kit

9 Best Practices for da Vinci Surgical System Implementation

Leadership

- Define the leadership structure within the da Vinci® operating room to reduce variability between cases
- Identify core team participants (including patient-side assistant and anesthesiologist), emphasizing the importance of standardization
- Designate each role's territories of responsibility

  Example:

  Scrub: Sets up the back table and drapes the patient cart

  Circulator: Responsible for getting the patient

  Anesthesiologist: Induces and intubates the patient

Communication

- Emphasize the importance of two-way verbal communication between OR staff and surgeon
- Review "read-back" technique

  Example:

  Surgeon: "Please remove the Monopolar Curved Scissors from arm 1 and replace with a large needle driver."

  Patient-side assistant: "Monopolar Curved Scissors coming out of arm 1. Large needle driver coming in."

  Surgeon: "Thank you."

Task assignment

- Each team member should be assigned a specific task to encourage task overlap
- Encourage team members to take initiative when they see a task that needs to be completed
- Emphasize that task overlap has the potential to maximize efficiency

Standardization

- Standardize the da Vinci system location, operating room layout, instrument trays, and core da Vinci team members to minimize variability between procedures
- Designate instrument and accessory selections for each procedure
- Confirm pick lists have been submitted to sterile processing

10 Da Vinci Case Walkthrough

Surgeon only

Port Placement	Docking	Intraoperative Use	Surgeon Console	Troubleshooting
Place ports	Deploy boom	Instrument and endoscope insertion	Instrument and endoscope manipulation	Arm management
Position patient	Position patient cart	Instrument and endoscope removal	Instrument activation	Fault management
	Targeting	Instrument and endoscope exchange	Dual console control	Conversion
	Dock remaining arms and spacing	Intraoperative cleaning		Shutdown



Da Vinci technology training pathway

Feedback

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